Lingchen Kong

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EDUCATION

Wuhan University Wuhan, China

M.S. in Space Probe and Information Processing Technology 09/2024–06/2027 (Expected)

Major: Radio Wave Propagation, Millimeter Wave SAR Imaging | GPA: 3.50/4.0

Wuhan University Wuhan, China

B.S. in Electronic Information Science and Technology 09/2020–06/2024

Major: Radio Wave Propagation and Antenna | GPA: 3.66/4.0

Core Courses: Advanced Mathematics, Linear Algebra, Engineering Stochastic Mathematics, Signals and Systems, Digital Signal Processing, Electromagnetic Field Theory, Analog/Digital Electronic Circuits, Mathematical Physics Methods

Dissertation: *Research on Key Techniques of Short-wave Time Difference of Arrival Localization Method*

RESEARCH EXPERIENCE

Short-Range Millimeter Wave SAR Imaging 07/2025–Present

School of Earth and Space Science and Technology, Wuhan University

Role: Leading Researcher

Contribution:

- Building mmWave SAR system based on TI mmWave sensor (IWR6843ISK), DCA1000EVM, and FUYU FMC4030 motion controller
- Developing a complete motion control to data acquisition to SAR imaging algorithm pipeline
- Modifying Range-Doppler Algorithm for 3D SAR imaging with enhanced resolution
- Utilizing an improved 2D capon estimator based on the random matrix method for super-resolution SAR imaging

Funding:

1. the National Key R&D program of China (2023YFB3905100)
2. National Key Research and Development Program of China (No. 2023YFA1009100)
3. NSFC grant U23B2016
4. National Natural Science Foundation of China grants 42104150, 42104151

Machine Learning-Based Nowcasting Method for Shortwave Maximum Usable Frequency 01/2025–Present

School of Earth and Space Science and Technology, Wuhan University

Role: Leading Researcher

Contribution:

- Collected MUF data from CMONOC (Crustal Movement Observation Network of China) in the North China region and space weather data (F10.7, SSN, Dst indices)
- Developed and compared 5 deep learning models, 4 machine learning models, and the ITU reference model. Particularly, I designed a novel deep learning model called 1DResNet-Transformer-Attention to estimate MUF.
- The model provides an intelligent solution for frequency selection in high-frequency communication systems

Funding:

1. The National Key R&D program of China (2023YFB3905100)
2. National Key Research and Development Program of China (No. 2023YFA1009100)
3. NSFC grant U23B2016
4. National Natural Science Foundation of China grants 42104150

Effect of the Ionosphere on Over-the-Horizon-Radar and Short Wave Propagation 06/2024–03/2025

Electronic Information School, Wuhan University

Sub-project: 3D ray tracing method for sky-wave propagation

Role: Research Assistant

Contribution:

- Utilizing the fourth-order Runge-Kutta method for 3D shortwave ray tracing under the numerical assimilation ionosphere model.
- Designed four apps based on MATLAB App Designer and the existing algorithms. The apps are for short-wave communication channel modeling, 3D ray tracing, ionosphere clutter suppression, and OTHR localization, respectively.
- Prepared defense presentation slides and drafted the final project completion report.

Passive Time-Difference-of-Arrival System for Shortwave Signals 11/2023–06/2024

Electronic Information School, Wuhan University

Role: Leading Researcher

Contribution:

- Assisted in building a TDOA system across the South and Central China regions
- Reconstructed a regional ionospheric parameters model based on Convolutional Neural Networks
- I first utilized the Quasi-Newton method to solve the TDOA localization equation to roughly localize the transmitter source, then I performed the grid-searching method by leveraging the ray tracing method based on the ionospheric parameters and quasi-parabolic ionosphere model for precise localization.

PUBLICATIONS

[1] L. Kong, T. Liu, M. Liu, H. Xu, W. Qiao, F. Zhang, and C. Zhou, “Fast and Robust 2-D Capon method for Millimeter-Wave SAR imaging,” *IEEE Trans. Aerosp. Electron. Syst.*, under review.

[2] L. Kong, X. Li, G. Xia, M. Liu and C. Zhou, "Machine-Learning-Based Regional Nowcasting Method for HF Maximum Usable Frequency Driven by Ionospheric Total Electron Content: A Case Study of North China," *IEEE Trans. Antennas Propag.*, doi: 10.1109/TAP.2026.3717336.

[3] L. Kong, T. Liu, C. Zhou, and Z. Zhao, “Time-difference-of-arrival localization analysis based on regional ionospheric high resolution reconstruction,” *Syst. Eng. Electron.*, vol. 47, no. 11, pp. 3521–3530, 2025. DOI: 10.12305/j.issn.1001-506X.2025.11.01. (in Chinese)

HONORS & AWARDS

- Second-Class Outstanding Freshmen Scholarship 2024
- Huang Zhangren Scholarship 2022
- First-Class Outstanding Student Scholarship 2022
- Second-Class Outstanding Student Scholarship 2021
- Outstanding Student Cadre 2021
- Merit Student (San Hao Xue Sheng) 2021, 2022
- Reviewer of *IEEE Transactions on Instrumentation and Measurement*

TECHNICAL SKILLS

Programming Languages: MATLAB (Proficient), Python (Proficient), C++ (Intermediate)

Technical Tools: LaTeX (Proficient), Microsoft Office Suite

Hardware Platforms: TI mmWave Sensors (IWR6843ISK), DCA1000EVM, Motion Control Systems

Research Areas: SAR Imaging, Signal Processing, Machine Learning, Ionospheric Physics, Radio Wave Propagation

Languages: Chinese (native), English (professional working proficiency)